

DOCUMENT CONVERSION

Srs

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Software Requirements Specification (SRS)

Order Control System (نظام إدارة الطلبات لمطعم سوشي)

1. Introduction

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements for the **Order Control System**, a web-based order management system designed for a sushi dark kitchen operating multiple brands. This document serves as the primary reference for developers, stakeholders, and project managers.

1.2 Scope

The system will replace the current manual Google Sheets workflow with a centralized web application. It will handle manual order entry by cashiers, order tracking, delivery management, daily expenses, and comprehensive reporting. The system manages operations for four brands (Flower Sushi, Mastery Sushi, Niwa Sushi, Tobiko Sushi) across various delivery platforms (Talabat, Menus, InstaShop, HarryApp, Phone).

1.3 Definitions and Acronyms

- **Dark Kitchen:** A restaurant that operates exclusively for delivery (no dine-in).
- **Brand:** A virtual restaurant brand operating from the same physical kitchen.
- **Platform:** Delivery aggregators or direct ordering channels.
- **Role:** System permissions level (Owner, Manager, Cashier).
- **AOV:** Average Order Value.

1.4 References

- `ENGINEERING_DIRECTIVES.md` (Engineering rules and constraints)
- `implementation_plan.md` (Approved implementation phases)

2. Overall Description

2.1 Product Perspective

The system is a standalone web application replacing fragmented spreadsheets. It integrates order entry, delivery dispatch, and financial tracking into a single unified flow. It acts as the core operational software for the kitchen.

2.2 Product Functions

- **Order Management:** Entry, status tracking, cancellation (with reasons), and discounts.
- **Menu Management:** Categories and products per brand.
- **Delivery Management:** Zones, fees, and driver assignment (own, app, or external).
- **Financial & Expense Tracking:** Expense logging and daily shift closing.
- **Reporting & Dashboards:** Analytics for sales, top items, and driver cash collection.
- **Auditing:** Comprehensive logs of all critical actions to prevent fraud.
- **Customer Database:** Customer history and order tracking.

2.3 User Classes and Characteristics

- **Cashier:** Fast-paced data entry from printed receipts. Needs quick UX, clear status transitions, and easy driver assignment. Restricted permissions.
- **Manager:** Oversees daily operations, approves discounts, manages menu updates, and monitors daily reports.
- **Owner:** Full system access. Needs high-level financial reports, audit logs, and user management capabilities.

2.4 Operating Environment

- Web-based system (accessible via modern browsers: Chrome, Safari, Edge).
- Deployed on Vercel (Frontend) and Supabase (PostgreSQL Database & Auth).
- Accessed via tablets and computers in the kitchen (Windows/macOS/Android/iOS).

2.5 Design and Implementation Constraints

- Tech Stack: Next.js + PostgreSQL (Supabase) + Prisma.
- Authentication: Supabase Auth (Email + Password).
- Architecture: Strict Separation of Concerns (Business logic in `src/lib/` or `src/services/`).
- Fail-Safe Policy: **No Hard Deletes**. Cancelled orders must require a reason.

2.6 Assumptions and Dependencies

- Orders are printed from platform tablets, and cashiers manually enter them into this system.
- Application commissions are calculated externally (P&L sheet) and are out of scope for this system.

3. Functional Requirements

FR-AUTH: Authentication & Authorization

ID	Description	Priority	Acceptance Criteria
FR-AUTH-01	System must authenticate users via Email + Password (Supabase Auth)	Must	Users can log in; failed attempts show error; unauthenticated users redirect to login
FR-AUTH-02	System must enforce RBAC at both API and UI levels (Owner/Manager/Cashier)	Must	API rejects unauthorized calls; UI hides inaccessible features
FR-AUTH-03	Session must persist across page refreshes	Must	User stays logged in until explicit logout or token expiry
FR-AUTH-04	System must support secure logout	Must	Session destroyed; redirect to login

FR-MENU: Menu Management

ID	Description	Priority	Acceptance Criteria
FR-MENU-01	Manager/Owner can create, edit, and deactivate Categories per Brand	Must	CRUD operations work; each Category belongs to one Brand
FR-MENU-02	Manager/Owner can create, edit, and deactivate Products within Categories	Must	Product has: Name, Price, Description, Category, isActive
FR-MENU-03	Products can be deactivated (hidden) but never deleted	Must	Deactivated products don't show in order form but remain in DB
FR-MENU-04	Categories can be reordered (sortOrder)	Should	Drag-and-drop or manual sort order
FR-MENU-05	Cashier cannot access menu management	Must	API rejects menu changes from Cashier role
FR-MENU-06	When creating an order, only products from the selected Brand appear	Must	Brand selection filters the product list

FR-ORD: Order Management (Core Module)

ID	Description	Priority	Acceptance Criteria
FR-ORD-01	Cashier can create a new order by selecting: Platform, Brand, Customer, Products, Payment Method, Zone	Mus t	Order saves with status "New"; subtotal + delivery fee auto-calculated
FR-ORD-02	Orders follow strict state transitions via State Machine	Mus t	State Machine in <code>src/lib/orderStateMachine.ts</code> validates; UI only shows valid next steps
FR-ORD-03	No Hard Delete — Orders can only be Cancelled with mandatory reason	Mus t	No delete endpoint exists; Cancel requires <code>cancelReason</code>
FR-ORD-04	Discounts require reason + Manager/Owner approval	Mus t	API rejects discount without <code>discountReason</code> + <code>discountApprovedBy</code>
FR-ORD-05	Every status change records a timestamp automatically	Mus t	<code>createdAt</code> , <code>confirmedAt</code> , <code>preparingAt</code> , <code>readyAt</code> , <code>outForDeliveryAt</code> , <code>deliveredAt</code> , <code>cancelledAt</code>
FR-ORD-06	Customer phone number auto-lookup: if customer exists, auto-fill name + address	Mus t	Typing phone triggers search; existing customer data populates form
FR-ORD-07	New customers are automatically created in Customer DB when order is saved	Mus t	First-time phone number creates new Customer record
FR-ORD-08	Order number is auto-generated (sequential per day or global)	Mus t	Unique, non-editable order number
FR-ORD-09	Order can be edited before status reaches "Preparing"	Sho uld	Editing after Preparing is blocked; edits logged in Audit
FR-ORD-10	Order list shows Live Orders (non-terminal statuses) with real-time filter	Mus t	Default view shows open orders; filter by status/platform/brand/date
FR-ORD-11	Order detail page shows full timeline (all timestamps + who did what)	Mus t	Visual timeline with timestamps and actor names
FR-ORD-12	When driver type is "App" (التطبيق يوصل بمنداييه), delivery fee = 0 automatically	Mus t	Selecting app driver sets fee to 0; no manual override needed

FR-DEL: Delivery Management

ID	Description	Priority	Acceptance Criteria
FR-DEL-01	Cashier can assign a driver when order status is "Ready"	Must	Assignment changes status to "Out for Delivery" + records driverId + driverAssignedAt
FR-DEL-02	System manages Delivery Zones with a single configurable fee per zone	Must	Fee auto-applies to order; Manager/Owner can edit zone fees at any time
FR-DEL-03	Drivers are categorized: OWN (مندوب خاص) / APP (مندوب التطبيق) / EXTERNAL (Indrive) / PICKUP (العميل)	Must	Driver type determines delivery fee behavior
FR-DEL-04	Driver cash collection is tracked per driver per day	Must	System knows how much cash each driver collected and must return
FR-DEL-05	Manager/Owner can add/edit/deactivate Drivers and Zones	Must	CRUD operations; deactivated items hidden from selection

FR-EXP: Expense Management

ID	Description	Priority	Acceptance Criteria
FR-EXP-01	System allows logging expenses with Type + Description + Quantity + Value + Date	Must	All fields required; date defaults to today
FR-EXP-02	20 predefined expense types seeded from current Excel	Must	Default types present on first use
FR-EXP-03	Manager/Owner can add new expense types	Must	New types appear in dropdown immediately
FR-EXP-04	Expense list filterable by type and date range	Should	Filter controls on expense list page

FR-CLOSE: Daily Closing

ID	Description	Priority	Acceptance Criteria
FR-CLOSE-01	Cashier can open a shift (records openedAt)	Must	Only one shift open at a time per cashier
FR-CLOSE-02	System auto-calculates closing summary	Must	Totals match raw order + expense data exactly
FR-CLOSE-03	Closing summary includes: total orders (success/cancel), cash, visa, driver cash, delivery fees, expenses, net cash	Must	All 7+ metrics calculated correctly
FR-CLOSE-04	Calculations happen in <code>src/lib/closing.ts</code> (Separation of Concerns)	Must	No business logic in page components
FR-CLOSE-05	Closing history viewable by date	Should	Past closings accessible with all details

FR-RPT: Reports & Dashboard

ID	Description	Priority	Acceptance Criteria
FR-RPT-01	Dashboard shows Live Orders + last 5 orders for Cashier	Must	Cashier sees only operational data
FR-RPT-02	Dashboard shows today's summary (orders, sales, cash/visa) for Manager/Owner	Must	Summary cards with key metrics
FR-RPT-03	Daily and Monthly reports with: orders, sales by platform+brand, delivery fees, expenses, net revenue, AOV	Must	All metrics accurate; filterable by date range
FR-RPT-04	Driver Cash Collection report	Must	Per-driver breakdown of cash orders and amounts
FR-RPT-05	Top Selling Products report	Must	Products ranked by quantity sold
FR-RPT-06	Reports exportable to PDF and Excel	Should	Export buttons generate downloadable files
FR-RPT-07	Backend enforced: API checks role before returning report data	Must	Cashier API calls to report endpoints return 403

FR-AUD: Audit Log

ID	Description	Priority	Acceptance Criteria
FR-AUD-01	Every Create/Update/Cancel on Order auto-logs to AuditLog	Must	Log entry has: userId, action, entityType, entityId, oldValue, newValue, timestamp
FR-AUD-02	Audit Log is immutable — entries cannot be edited or deleted	Must	No UPDATE/DELETE endpoints for AuditLog table
FR-AUD-03	Audit Log viewable by Owner only	Must	API returns 403 for non-Owner; UI hides link
FR-AUD-04	Audit Log filterable by: user, action type, date range	Should	Filter controls on audit page
FR-AUD-05	Audit logging triggered automatically via centralized function, not manually per page	Must	All DB writes route through <code>src/lib/audit.ts</code>

FR-USR: User Management

ID	Description	Priority	Acceptance Criteria
FR-USR-01	Owner can create new users (name + email + password + role)	Must	New user can log in immediately
FR-USR-02	Owner can edit user details and change roles	Must	Role change takes effect on next login/refresh
FR-USR-03	Owner can deactivate users (not delete)	Must	Deactivated user cannot log in
FR-USR-04	Only Owner can access User Management	Must	API + UI enforced

FR-CUST: Customer Database

ID	Description	Priority	Acceptance Criteria
FR-CUST-01	Customer identified by unique phone number	Must	Duplicate phones rejected
FR-CUST-02	Auto-create customer on first order	Must	New phone → new Customer record with name + address from order
FR-CUST-03	Auto-fill customer data when phone matches existing record	Must	Phone input triggers lookup; name + address populate
FR-CUST-04	Customer profile page shows: basic info, total orders, last order date, order history	Must	All data visible on customer detail page
FR-CUST-05	Customer profile shows problem orders : cancelled, quality issues, delivery problems	Must	Filterable list of problematic orders
FR-CUST-06	Customer profile shows whether they are "first-time" or "returning"	Must	Badge or indicator on customer card
FR-CUST-07	Customer list searchable by name or phone	Must	Search bar with instant results
FR-CUST-08	Customer notes field (free text for special instructions or issues)	Should	Editable notes visible to all roles

4. Data Requirements

- **User**: ID, Email, Name, Role
- **Brand**: ID, Name, IsActive
- **Product/Category**: Catalog data linked to Brands
- **Customer**: ID, Name, Phone (Unique), Address, Notes, Order History
- **DeliveryZone**: ID, Name, Fee
- **Order**: Status, PaymentMethod, Financials (Subtotal, Discount, DeliveryFee), Timestamps
- **AuditLog**: Automatically populated on DB writes.

Note: All schema changes must be documented in `PROJECT_LOG.md` immediately upon creation.

5. External Interface Requirements

5.1 User Interfaces

- **Languages:** Bilingual (Arabic & English) via `next-intl`.
- **Layout:** Full RTL/LTR support.
- **Responsiveness:** Must work on tablets (for kitchen use) and desktop monitors.

5.2 Hardware Interfaces

- No direct hardware integration. The system will be used alongside existing tablet and receipt printer setups. Cashiers transcribe from printed receipts.

5.3 Software Interfaces

- **Supabase Auth:** For user identity and session management.
 - **PostgreSQL:** Primary data store.
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6. Open Items

- Finalizing the exact Menu Structure (Categories/Products/Variants) pending receipt samples from the client.
- Verifying if any specific custom export formats are strictly required beyond standard Excel/PDF.